

HENRY BORY PREVEZ

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Update: January 27, 2025



EDUCATION:

University of Oriente

Dr. C. in Automation

University of Oriente

Master of Science in Automation- GPA 5.00/5.00

University of Oriente

Graduated of Automatic Engineering - GPA 4.97/5.00

Santiago de Cuba, Cuba

Expected November 2021

Santiago de Cuba, Cuba

Expected March 2011

Santiago de Cuba, Cuba

Expected July 2002

SKILLS:

- MatLab/Simulink, PSim, Python, MS Office Suite, Statgraphic.
- Differential and Integral Calculus, Mathematical Analysis, Numerical Analysis, Descriptive and Inferential Statistics.
- Fluent Spanish and English.
- Systems Modeling, Simulation and Identification.
- Digital Signal Processing, FFT and filter design.
- Systems Dynamics
- Experimental Statistic Design.
- Power electronics.
- Electric Circuits.
- Analog and Digital Control System.
- Advanced Control Methods.
- Linear and Nonlinear Control Systems Design and System Identification.
- Computer aided control system design.
- Fuzzy Control.
- Industrial Automation at the beginning of Industry 4.0.
- Model based Predictive Control (MPC) design of linear time invariant systems.

INTERNATIONAL RELEVANT PROJECT:

CRECE (Capacity Building for Renewable Energy in Cuban Higher Education)

September 2017 – 2020

- Propose the combination of the symmetrical switching form with two rectifier topologies, the single phase rectifier to diodes with switch in series with the load and the three phase to diodes with switch in series with the load.

IRIS (Integrating of Renewable Intermittent Source)

September 2019 – 2022

- Propose the calculus Methodology of the dump load and the components for two rectifier topologies.
- Design of the frequency regulator and the demonstration of the insensibility of its parameters respect to the rectifier topology employed in the loop frequency regulation.

EXPERIENCE:

Professor of the University of Oriente

September 2002 – Nowadays

- Instructor of course: Electric Machineries, Electrics Drives, Power Electronics and Control Theory.
- Software register: TabPer V 1.0, AnaRed V1.0, Cal_Arr_Fre_V1.0, Calc_Covid19 V. 1.0 and Niord v. 1.0.
- Wrote and registered the monograph: An actualization of the friction, its compensation and control method in electromechanics systems.
- Coordinator of the International Project IRIS, representing the University of Oriente.

PUBLICATIONS OF THE LAST FIVE YEARS:

Bory H., et al. AnaReg_V1 as MatLab aid for analyzing of regulation systems and controller tuning in Instrumentation and Controls. Tecnología Química, No 1, Vol.44, pp. 122-136. enero-abril 2024. ISSN: 2224-6185.

http://scielo.sld.cu/scielo.php?script=sci_arttext&pid=S2224-61852024000100122&lang=es

Bory H., et. al. Laboratory practice of Power Electronic using the mobile application EveryCircuit. vol. 13. 2024. ISSN-e: 2310-3637. <https://www.horizontepedagogico.cu/index.php/hop/article/view/352/733>

Bory H., et. al. Proposal for the VirtualLab BR4GL educational software to support the teaching of Robotics. No. 1, vol. 13. 2024. ISSN-e: 2310-3637. <https://www.horizontepedagogico.cu/index.php/hop/article/view/349/716>

Bory H., et al. MatLab model of tidal thermal power plants based on Anderson cycle. First approach. INGENIERÍA HIDRÁULICA AMBIENTAL. No. 01, Vol. XLIV, pp. 77-90. Ene-Mar 2024. ISSN 2788-6050. <http://scielo.sld.cu/pdf/riha/v45n1/1680-0338-riha-45-01-77.pdf>

Bory H., et al. Proposal for a virtual laboratory for the Ecology subject of the Bachelor's degree in Biology. Revista Santiago, No 160, pp. 298-311. 2023. e-ISSN: 2215-4812.

<https://santiago.uo.edu.cu/index.php/stgo/article/view/5537/5018>

Bory H., et al. Comparison of three tidal power plant schemes. MAESTRO y SOCIEDAD. No. 4, Vol. 20, pp. 953-958, 2023. e-ISSN: 1815-4867. <https://maestroysociedad.uo.edu.cu>

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Bory H., et al. Simulations of surface charge density changes during the untreated solid tumour growth. Revista Royal Society Open Science. Vol. 9, No. 11. 2022. <https://royalsocietypublishing.org/doi/10.1098/rsos.220552>

Bory H., et al. Time behaviors of the active and accumulative cases during the second and third COVID-19 epidemic reappearances for Santiago de Cuba province. Anales de la Academia de Ciencias de Cuba, Vol. 12, No. 1. 2022. ISSN 2304-0106.

Bory H., et al. Evaluation of the Symmetrically Switched Converter Structures on the Frequency Regulation of Standalone Micro Hydro Power Plants. 11 th IFAC Symposium on Control of Power and Energy Systems (CPES 2022). <https://doi.org/10.1016/j.ifacol.2022.07.013>

Bory H., et al. Effect of Symmetrically Switched Rectifier Topologies on the Frequency Regulation of Standalone Micro-Hydro Power Plants. Energies, 14. 2021. <https://doi.org/10.3390/en14113201>

Bory H., et al. Symmetrical Angle Switched Single-Phase and Three-Phase Rectifiers: Application to Micro Hydro Power Plants. ScienceDirect, Vol. 52, No. 4, pp. 216-221. 2019.

Bory H., et al. Comparison between single phase Rectifier with Symmetrical Switching and AC/AC converter to Improve the Power Factor in Micro Hydro Power Plants. RIAI. Vol. 1, No. 6, pp. 79-88, 2019. <https://doi.org/10.4995/riai.2018.9313>

Bory H., et al. Comparison between three phase Rectifier with Symmetrical Switching and AC/AC converter to Improve the Power Factor in Micro Hydro Power Plants. RIAI. Vol. 1, No. 5, pp. 101-111, 2018. <https://doi.org/10.4995/riai.2017.8816>